

Slide 1

Hello my name is

Slide 2

"RFID, or Radio-Frequency Identification, is a technology that enables wireless communication between a reader and a tag using radio waves. This is commonly used for authentication in access control systems, payment cards, and IoT devices. The system consists of three main components:

1. **RFID Tag** – Contains data stored on a microchip and can be passive (powered by the reader) or active (has its own battery).
2. **RFID Reader** – Transmits radio signals to the tag and retrieves its stored information.
3. **Backend System** – Processes the received data and verifies authenticity.

Since RFID operates wirelessly, it is susceptible to various security threats."

(Why is RFID Security a Concern?)

"RFID tags often lack strong encryption due to hardware limitations. Unlike traditional security, RFID devices must prioritize efficiency and low power consumption. This makes them vulnerable to:

- **Eavesdropping** – Attackers intercept RFID signals and extract sensitive data.
- **Cloning** – An attacker copies the tag's information and replicates it on a blank RFID card.
- **Relay Attacks** – Extending the communication range to trick the system into granting access.
- **Side-Channel Attacks** – Extracting information by analyzing physical characteristics like power consumption or electromagnetic emissions.

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(What is Side-Channel Analysis?)

"Side-channel attacks do not target the encryption algorithm itself but rather the unintended data leaks during computation. These leaks can come from:

- **Power Consumption** – Measuring fluctuations when the RFID chip processes data.
- **Electromagnetic Emissions** – Capturing signals emitted by the RFID circuit.
- **Timing Analysis** – Observing variations in execution time to infer information.

These attacks can be extremely effective, even against encrypted RFID systems."

(Why is Side-Channel Analysis a Threat to RFID?)

"RFID devices are particularly vulnerable because they operate in open environments with limited security mechanisms. Attackers can extract data by simply being near the target.

In this presentation, I will discuss several studies that analyze RFID vulnerabilities and demonstrate how side-channel attacks can be used to extract sensitive data."

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These studies build on each other by progressively exploring different methods of attacking and analyzing RFID security. Let's go through them one by one.

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Key Idea:

"This study focused on cloning RFID cards by intercepting the data and writing it onto a blank RFID card.

How was it done?

- Researchers used an RFID reader to capture authentication data.
- They then transferred this data onto another RFID card.
- The cloned card was successfully used to gain unauthorized access.

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- Bit about the images

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- How apps can interpret the data

Significance:

This proved that RFID systems without strong encryption can be easily cloned, highlighting the importance of secure key management and authentication mechanisms."

(How does this connect to later studies?)

"This study demonstrates a simple yet effective attack. The next study goes a step further by attempting to reverse-engineer RFID firmware to extract even more insights."

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Key Idea:

"Instead of simply cloning RFID cards, this study investigated RFID firmware to understand how the system processes authentication."

How was it done?

- The researchers analyzed firmware used in RFID access systems.
- They studied how the software interacted with RFID tags.
- They found weaknesses in how authentication data was processed.

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- Talk about the methodology
- Talk about the image

This image represents intercepted RFID communication between a tag and a reader, specifically showing raw data packets exchanged during authentication or another RFID process

Significance:

- Even though no critical vulnerability was found, the study highlighted privacy concerns.
- It showed that with enough effort, attackers could reconstruct parts of the RFID authentication process."

(How does this connect to the previous study?)

"This builds on the cloning attack by showing that attackers can extract even more information from RFID systems by analyzing the internal software. However, what if an

attacker doesn't need to reverse-engineer firmware but can instead analyze power consumption to extract data? That's where the next study comes in."

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Key Idea:

"This study demonstrated that machine learning models could be trained to recognize power consumption patterns and extract encryption keys from RFID systems."

How was it done?

- Researchers collected power traces from an RFID system during authentication.
- They used machine learning algorithms to find patterns in the power consumption.
- The model was able to predict encryption keys with high accuracy."

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- Talk about images

Correlation Between Traces and Leakage Model (16 Uniform Classes)

- This shows how well the leakage model aligns with actual power traces.

Correlation Between First and All Instances (16 Uniform Classes)

- Evaluates consistency across multiple measurements.

Correlation Between Traces and Leakage Model (9 Binomial Classes)

- Uses a different leakage model (9 binomial classes instead of 16 uniform).

Fig. 6: Correlation Between First and All Instances (9 Binomial Classes)

- Tests how stable the leakage is across runs for this model.
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Higher noise reduces correlation, making side-channel attacks less effective.

Both leakage models (16 uniform and 9 binomial classes) show similar trends—increasing noise disrupts correlation.

Significance:

- This study introduced a much more advanced and scalable method of attacking RFID security.
- It proved that side-channel attacks are not just theoretical—they can be automated and optimized using AI.

(How does this connect to previous studies?)

"This study builds on the previous research by showing that attackers don't even need direct access to RFID software. Instead, they can use AI to analyze physical characteristics like power consumption. This represents a significant escalation in the sophistication of RFID attacks."

Slide 12 – Proposed solution

Slide 13 – Plan

Slide 14 – Conclusion

RFID security is a growing concern.

Side-channel attacks are becoming more advanced.

Effective countermeasures must be developed to protect future RFID systems.

Slide 15 – Questions

Any questions

Slide 16 – Thank you

Thank you

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